Snack Health Risk Analyzer

Detecting Harmful Chemicals in Packaged Snack Foods Using NLP, Machine Learning, and the Comparative Toxicogenomics Database

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Abstract

This project started with a simple but troubling observation. A popular Korean snack product, Buldak ramen, carries absolutely no health warnings on its packaging when sold in South Korea. The exact same product sold in the United States carries a bold California Proposition 65 warning that reads: 'WARNING: Cancer and Reproductive Harm.' Same food, same ingredients, two completely different levels of consumer information — depending entirely on which country you happen to live in. That disparity raised a question we couldn't ignore: if harmful chemicals in food are real enough to trigger legal warnings in one country, why do most consumers around the world have no way of knowing they are eating those same chemicals?

The Snack Health Risk Analyzer was built to answer that question at scale. The system processes 99,958 real snack food products from the OpenFoodFacts global database, automatically detects harmful chemical additives using Natural Language Processing, calculates a scientifically-grounded risk score based on WHO, FDA, and IARC evidence, maps each detected chemical to its associated diseases using the Comparative Toxicogenomics Database, and classifies every product as Safe, Low, Medium, or High risk.

Two parallel pipelines were developed. Pipeline 1, the full production system, runs on all 100,000 products and uses a manually-built database of 207 harmful chemicals with weights directly derived from regulatory decisions and peer-reviewed research. Pipeline 2, the R&D system, applies multilingual NLP including language detection, encoding repair, and machine translation to a 5,000-product sample, then integrates with CTD's 179,459-chemical database to assign risk weights automatically from InferenceScores — no manual input required. The final Random Forest classifier achieved 99% test accuracy with an F1-weighted score of 0.9950. Results show that 31.1% of analyzed snacks contain at least one harmful ingredient, with cancer risk, digestive issues, and heart disease being the most common associated conditions.

1. Introduction

1.1 The Story That Started This Project

In 2024, a photograph of a Buldak ramen product began circulating online. The image showed two versions of the same product side by side — the South Korean version with no health warnings at all, and the US version with a California Proposition 65 label warning consumers about cancer and reproductive businesses to provide warnings before knowingly and intentionally exposing anyone to chemicals that cause cancer, birth defects, or other reproductive harm. The warning on Buldak was triggered by the presence of certain chemical additives in the product.

What made this image so striking was not just the warning itself — it was the contrast. In Korea, no such warning exists, even though both countries' consumers are eating the same product. This is not a regulatory inconsistency unique to one product or one country. It reflects a global reality: the same chemical additives appear in food products across the world, but the information available to consumers varies dramatically based on geography, regulatory environment, and the language of the product label. A consumer in France, Germany, or Japan eating a snack with TBHQ, sodium benzoate, or partially hydrogenated oils has no automated, accessible way to know what those chemicals do to the human body.

This observation became the founding motivation for the Snack Health Risk Analyzer. If harmful chemicals are real enough to warrant legal warnings in California, they are real enough to be detected, scored, and communicated to every consumer — regardless of where they live or what language their food labels are written in.

1.2 The Scale of the Problem

Packaged snack foods are one of the largest categories in the global food market. Chips, cookies, chocolate bars, crackers, candy, and baked snacks are consumed daily by billions of people. They are also the category most heavily loaded with artificial additives — preservatives to extend shelf life, artificial colors to make products visually appealing, emulsifiers to blend oil and water, flavor enhancers to intensify taste, and sweeteners to reduce sugar while maintaining sweetness. Many of these additives are synthetic, petroleum-derived, or coal tar-derived chemicals that have no nutritional value and exist solely to serve manufacturing and commercial purposes.

A typical packaged snack product contains between 15 and 45 ingredients. Reading an ingredient list and accurately identifying which ingredients are harmful, why they are harmful, and what diseases they are linked to requires specialist chemical knowledge that most consumers do not have. Even those who try to research specific ingredients face a fragmented information landscape — scientific evidence is scattered across regulatory agency websites, academic journals, and toxicology databases that are not designed for public use.

The OpenFoodFacts database, the largest open-source food product database in the world, contains data on over 2.7 million food products from 174 countries. The ingredient lists in this database are written in over 20 languages and use a mix of chemical names, E-numbers, brand names, and abbreviated codes to describe the same substances. No automated system currently exists that reads this multilingual ingredient data, identifies the harmful chemicals, and tells a consumer how dangerous a product actually is — based on real scientific evidence.

1.3 Problem Statement and Objectives

The core problem this project solves is the following: given the raw, unstructured, multilingual ingredient text from any packaged snack product, can we automatically determine which harmful chemical additives it contains, quantify the severity of the risk those chemicals represent, identify what diseases those chemicals are linked to, and classify the product into a meaningful risk category — all without any human review?

The specific objectives of this project are:

- Build an automated NLP pipeline that detects chemical additive names in raw multilingual ingredient text regardless of the language they are written in
- Create a scientifically-grounded risk scoring system where every chemical's risk weight is directly traceable to a specific regulatory decision or peer-reviewed study
- Integrate the Comparative Toxicogenomics Database (CTD) to automatically link detected chemicals to their peer-reviewed disease associations
- Engineer meaningful machine learning features from chemical detection results and train a classifier to predict product risk level
- Handle encoding corruption, language detection, and machine translation for multilingual ingredient text

- Evaluate model performance using class-imbalance-aware metrics and compare four machine learning algorithms
- Compare a manually-curated detection approach with a fully automated CTD-driven pipeline

1.4 Scope and Structure

This project focuses on snack food products — the category where harmful additive usage is most concentrated. The system processes 99,958 products from OpenFoodFacts. The chemical database covers 207 confirmed harmful substances across seven categories. The R&D pipeline was applied to a 5,000-product multilingual sample. All code is written in Python using open-source libraries.

Section 2 surveys the literature and existing tools. Section 3 describes the datasets. Section 4 covers Exploratory Data Analysis with visualization placements. Section 5 details the complete methodology for both pipelines. Section 6 presents results and evaluation. Section 7 discusses implications and challenges. Section 8 concludes with future directions.

2. Literature Review

2.1 Food Additive Safety — The Regulatory Landscape

Food additive safety regulation is not uniform across the world. In the United States, the FDA operates under a Generally Recognized as Safe (GRAS) framework — a chemical can be used in food unless there is sufficient evidence of harm to justify restriction or removal. In the European Union, the EFSA operates under the precautionary principle — a chemical must be evaluated and approved before it can be used. This fundamental philosophical difference means that many chemicals permitted in US food products are either banned or require warning labels in the EU, and vice versa. The Buldak case is one visible example of this regulatory asymmetry.

The Joint FAO/WHO Expert Committee on Food Additives (JECFA) establishes Acceptable Daily Intake values for food additives globally, but these values are calculated for individual chemicals in isolation. The cumulative exposure to multiple additives consumed simultaneously — which is the reality for any processed snack product — is rarely evaluated (Boobis et al., 2008). A person eating a bag of flavored chips might consume four or five harmful chemicals at once. No regulatory framework adequately addresses this cumulative burden, and no existing consumer tool quantifies it. This project does.

2.2 Key Studies That Informed Our Risk Weights

Every risk weight in our chemical database is traceable to a specific study or regulatory decision. The following are the most important:

Southampton Study (McCann et al., 2007) — Weight basis for artificial colors (W:15): A randomized, double-blind, placebo-controlled trial conducted at the University of Southampton and published in The Lancet demonstrated that a mixture of six artificial food dyes — Tartrazine (E102), Sunset Yellow (E110), Carmoisine (E122), Allura Red (E129), Quinoline Yellow (E104), and Brilliant Blue (E133) — combined with sodium benzoate caused statistically significant increases in hyperactivity and attention deficit behaviors in children aged 3 and 8–9. This was a landmark study because it was the first controlled trial at realistic dietary doses to confirm what parents had suspected for years. The EU responded by requiring warning labels on all products containing these six dyes. This study is the direct scientific basis for the weight of 15 assigned to artificial color chemicals in our database. Source: The Lancet, https://doi.org/10.1016/S0140-6736(07)61306-3

IARC Monographs Volume 114 (Bouvard et al., 2015) — Weight basis for sodium nitrite (W:20): The International Agency for Research on Cancer classified processed meats as Group 1 carcinogens — meaning there is sufficient evidence that they cause cancer in humans. The primary mechanism is the formation of nitrosamines when sodium nitrite (E250) reacts with secondary amines under the acidic conditions of the stomach. Nitrosamines are among the most potent known carcinogens, with strong links to colorectal, stomach, and pancreatic cancer. This Group 1 classification is the second-strongest possible signal from the world's leading cancer research agency, and it justified a weight of 20 for sodium nitrite in our system. Source: The Lancet Oncology, https://doi.org/10.1016/S1470-2045(15)00444-1

FDA Final Determination on Partially Hydrogenated Oils (FDA, 2018) — Weight basis for trans fats (W:25): The U.S. Food and Drug Administration revoked GRAS status for partially hydrogenated oils in 2018, completing a multi-year process that began with the FDA's preliminary determination in 2013. The decision was based on overwhelming evidence that industrially-produced trans fatty acids raise LDL cholesterol, lower HDL cholesterol, promote systemic inflammation, and are causally linked to coronary heart disease. The FDA estimated that removing partially hydrogenated oils from the food supply would prevent approximately 20,000 heart attacks and 7,000 deaths annually in the United States. A complete government ban — the strongest possible regulatory action — justified the highest weight of 25 in our database. Source: FDA website, https://www.fda.gov/food/food-additives-petitions/trans-fat

IARC Monographs Volume 134 — Aspartame (IARC, 2023) — Weight basis for aspartame (W:12): In July 2023, IARC classified aspartame (E951) as Group 2B — possibly carcinogenic to humans. This classification was controversial because it was based primarily on limited human evidence from observational studies and mechanistic evidence from in vitro research. Nevertheless, it represented a meaningful regulatory signal from the world's leading cancer classification authority. Aspartame breaks down in the body to phenylalanine, methanol, and aspartic acid — metabolites with documented neurological effects at high doses. The 2023 classification justified a weight of 12. Source: IARC, https://www.iarc.who.int/featured-news/iarc-monographs-evaluate-the-carcinogenicity-of-aspartame/

Chassaing et al. (2015), Nature — Weight basis for emulsifiers (W:8–10): A landmark study published in Nature demonstrated that two common food emulsifiers — carboxymethylcellulose (E466) and polysorbate 80 (E433) — disrupted gut microbiota, increased intestinal permeability and inflammation, and promoted colitis and metabolic syndrome in mice at doses approximating normal human consumption. What made this study particularly significant was that these effects were observed below

regulatory ADI limits — suggesting current safety standards may be insufficient. Follow-up work by Viennois et al. (2017) showed these emulsifiers promoted colon cancer development. This evidence justified weights of 8–12 for emulsifier chemicals. Source: Nature, https://doi.org/10.1038/nature14232

EFSA Re-evaluation of Titanium Dioxide / EU Ban 2022 (EFSA, 2021) — Weight basis for E171 (W:15): The European Food Safety Authority conducted a comprehensive re-evaluation of titanium dioxide (E171) and concluded it could no longer be considered safe as a food additive. The primary concern was genotoxicity — the ability to damage DNA. Commercial E171 contains a significant fraction of nanoparticles with at least one dimension below 100 nanometers, which can cross the gut barrier and accumulate in organs. Bettini et al. (2017) demonstrated precancerous lesions in animals after oral E171 exposure. The EU banned E171 as a food additive in 2022 — the only food additive to be banned by the EU on genotoxicity grounds in decades. This justified a weight of 15. Source: EFSA Journal, https://doi.org/10.2903/j.efsa.2021.6585

2.3 The Comparative Toxicogenomics Database

The Comparative Toxicogenomics Database (Davis et al., 2023) is the most comprehensive resource for understanding chemical-disease relationships from peer-reviewed literature. CTD is maintained by Mount Desert Island Biological Laboratory and is updated continuously. It contains over 179,000 chemicals, 10,000+ diseases, and millions of gene interaction records. CTD distinguishes between two types of evidence: curated direct evidence (where a trained curator manually reads a specific paper and annotates a chemical-disease relationship) and inferred relationships (where a chemical-disease connection is computed through shared gene interaction pathways).

The InferenceScore — a metric unique to CTD — quantifies the strength of inferred associations. It is calculated by finding all gene pathways connecting a chemical to a disease, multiplying the chemical-gene interaction strength by the gene-disease association strength for each pathway, then summing across all connecting pathways. A higher score means more gene pathways with stronger evidence. No existing food safety application integrates CTD InferenceScores as risk weights. Our R&D pipeline is, to our knowledge, the first to do this. Source: CTD database, https://ctdbase.org/

2.4 Existing Tools and Their Gaps

Several consumer food safety tools exist but all have significant limitations:

Tool	Coverage	Algorithm	Disease Links	Multilingual	Transparent
Yuka	EU/US products	Proprietary — not disclosed	None	French/English	No
EWG Food Scores	~80K US products	Manual curation	None	English only	Partial
Fooducate	US products	Nutritional grading	None	English only	No
Open Food Facts	2.7M products (global)	None — raw data only	None	20+ languages	Yes
This Project	99,958 snacks (global)	NLP + ML + CTD evidence	CTD-backed disease links	20+ languages	Yes — every weight cited

The key gap is clear: no existing tool connects ingredient chemical detection to peer-reviewed disease data, explains its scoring methodology, or handles multilingual ingredient text at global scale. This project fills all three gaps simultaneously.

3. Data Description

3.1 Primary Dataset — OpenFoodFacts

The primary dataset was sourced from Open Food Facts (https://world.openfoodfacts.org/data), the world's largest open-source food product database. The raw source file is en.openfoodfacts.org.products.csv — a tab-separated file exceeding 12 gigabytes containing over 2.7 million products from 174 countries. Because loading a 12GB file into memory at once would crash most machines, a chunk-based processing approach was used: the file was read 50,000 rows at a time using pandas, with snack-relevant rows filtered at each chunk before storing. This is a standard big data technique for handling files that exceed available RAM.

Products were filtered using keyword matching on the categories_en column. The following snack-related keywords were used: chips, crisps, cookies, biscuits, cake, chocolate, candy, bar, cracker, snack. After filtering, duplicate rows were removed by dropping products with identical product_name and ingredients_text combinations. A random sample of 100,000 products was then drawn using random_state=42 to ensure reproducibility. After removing rows with missing ingredient text, the final dataset contained 99,958 usable rows.

3.2 Original Dataset Columns — What Was Kept and Why

The OpenFoodFacts source file contains over 180 columns. Most were irrelevant to ingredient-based chemical analysis and were dropped. The table below explains every decision:

Column	Decision	Reason
product_name	Kept	Product identification
categories_en	Kept	Category filtering; used for context
ingredients_text	Kept	Raw multilingual ingredient list — primary analysis target
ingredients_cleaned	Kept	Pre-cleaned version with encoding fixes applied
brands	Dropped	Not relevant to ingredient chemistry
quantity / serving_size	Dropped	Dosage analysis is out of scope for this version
energy_100g / fat_100g / sugars_100g / proteins_100g / salt_100g	Dropped	Nutritional macros out of scope — this project targets additives not nutrition

additives_tags	Dropped	Extremely sparse — most products with harmful additives have no entries here; using raw ingredients_text gives better coverage
countries_en / labels_en / created_datetime	Dropped	Not relevant to chemical risk analysis

The decision to drop additives_tags deserves emphasis. This column appeared to provide pre-parsed additive information, which would seem useful. However, an inspection revealed that most products with harmful additives visible in ingredients_text had no corresponding entry in additives_tags. Using it would have created severe selection bias — only well-documented products would appear to have additives. Processing the raw ingredients_text through our NLP pipeline ensures every product is treated consistently.

3.3 CTD Supporting Datasets

CTD Chemicals File (ctd_chemicals_sample.csv) — 179,459 rows: Contains ChemicalName, ChemicalID in MESH:XXXXXXX format, MESHSynonyms (pipe-separated), and CTDCuratedSynonyms (pipe-separated). This file was used to build a comprehensive synonym dictionary — indexing every possible name variant for each chemical — enabling detection even when labels use unusual naming conventions. Source: https://ctdbase.org/downloads/

CTD Disease File (ctd_sample.csv) — 5,000 rows: Contains ChemicalName, ChemicalID (without MESH: prefix), DiseaseName, DirectEvidence, InferenceScore, and PubMedIDs. This provides the chemical-to-disease mapping for the R&D pipeline.

Critical Bug Found: The two CTD files used inconsistent ChemicalID formats. The chemicals file used 'MESH:C046983' while the disease file used 'C046983'. This caused all join operations to return zero matches. The fix was one line:

```
df_chem["ChemicalID_clean"] = df_chem["ChemicalID"].str.replace("MESH:", "",
regex=False).str.strip()
```

After this fix, 2,458 unique chemicals with confirmed disease associations were successfully matched between the two files.

3.4 Dataset Limitations and Honest Caveats

- OpenFoodFacts is crowdsourced — product entries are contributed by volunteers and may contain incomplete or inaccurate ingredient information

- Ingredient text is unstructured — there is no standard format across products, countries, or languages
- No quantity information — we detect the presence of chemicals but cannot assess dosage, which determines actual health impact
- CTD sample file has pharmaceutical bias — food additive coverage is lower than pharmaceutical compound coverage, limiting Pipeline 2's detection rate
- Name aliases vary — 'red 40 lake' and 'allura red' are the same chemical but written differently on different products; our database catches most but not all variants

4. Exploratory Data Analysis

4.1 Language Distribution — The Multilingual Problem

The first and most significant finding during EDA was the scale of the multilingual challenge. Running langdetect on the 5,000-product R&D sample revealed that only 40.6% of ingredient texts were written in English. The remaining 59.4% spanned over 20 languages:

Language	Products	Percentage	Notes
English	2,029	40.6%	Primary detection language
French	1,528	30.6%	Largest non-English group — many EU products
German	345	6.9%	German market products
Spanish	273	5.5%	Spanish/Latin American products
Italian	185	3.7%	Italian market products
Dutch	103	2.1%	Netherlands products
Others (20+ languages)	537	10.7%	Portuguese, Norwegian, Polish, Romanian, etc.

This finding is critical: a system that only processes English text would miss harmful chemicals in 6 out of every 10 products. This motivated the entire NLP translation pipeline in Pipeline 2 and the E-number matching strategy used in both pipelines.

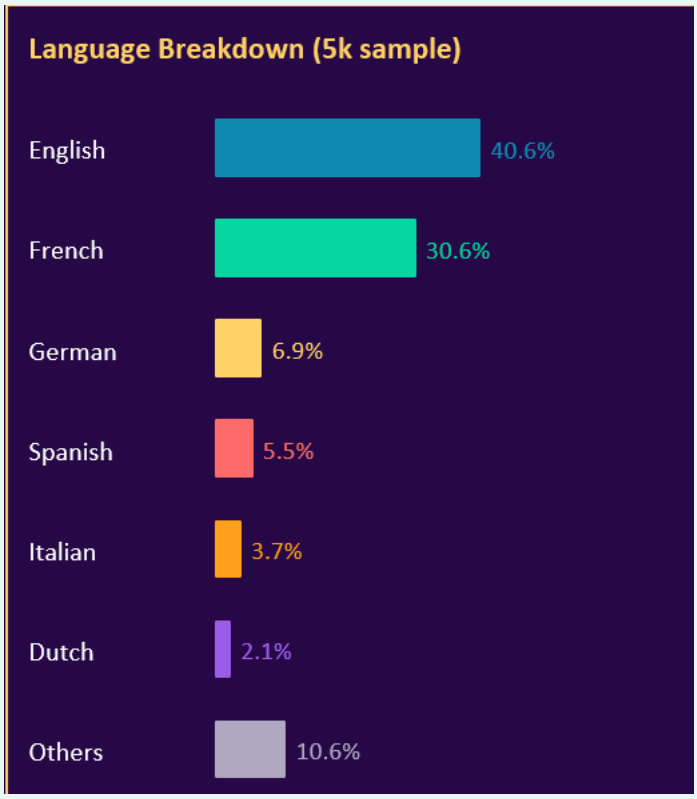

Figure 1: Language Distribution Bar Chart — from R_D.ipynb (language_distribution.png or inline chart)

4.2 Risk Level Distribution After Detection

After running the full chemical detection and risk scoring pipeline on 99,958 products, the distribution of risk levels across the dataset was as follows:

Risk Level	Products	Percentage	Score Range	Meaning
Safe	68,826	68.9%	0	No harmful ingredients detected
Low	20,908	20.9%	1–30	Minor additives — low individual risk
Medium	4,869	4.9%	31–60	Multiple concerning chemicals
High	5,355	5.4%	61–100	Severely harmful combination

The 68.9% Safe rate does not mean most snacks are healthy — it means most snacks do not contain the specific 207 chemicals in our database at detectable levels in their ingredient text. Products classified as Safe may still contain sodium, added sugars, and saturated fats in excess. However, 31.1% of products —

nearly 1 in 3 snacks — do contain at least one harmful chemical from our detection database, which is a significant public health finding.

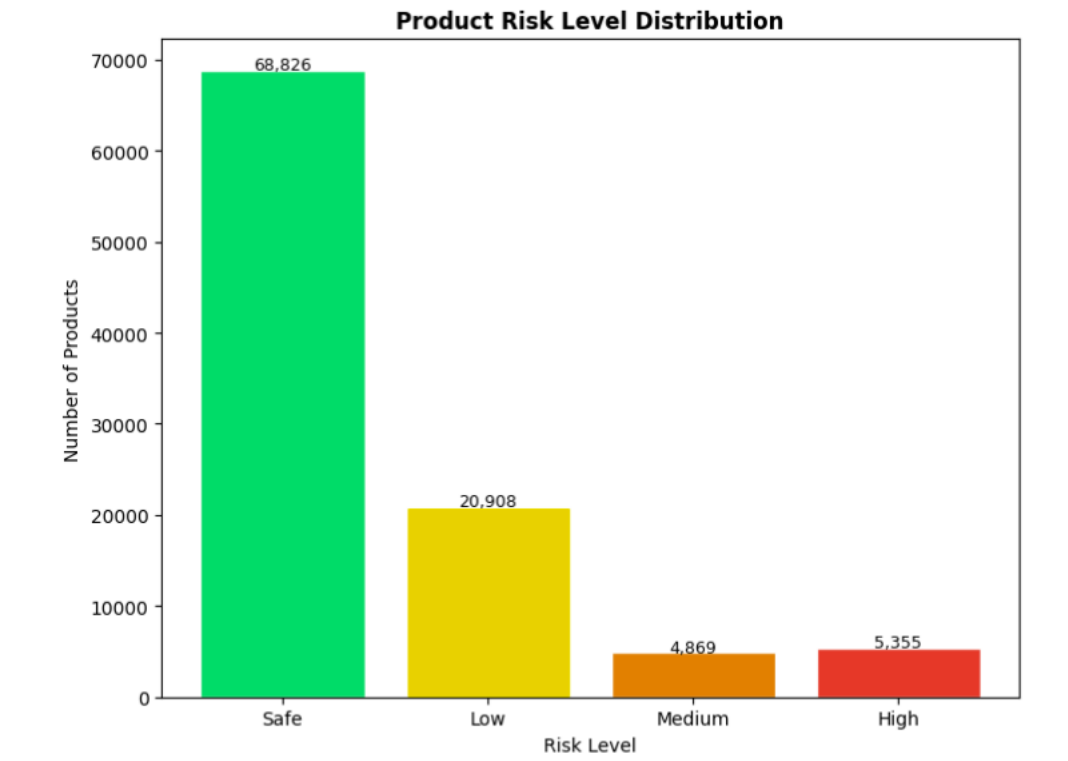

Figure 2: Risk Level Distribution Bar Chart — from Final_Version.ipynb (health_risk_analysis.png — top-left subplot)

4.3 Risk Score Distribution

The distribution of numerical risk scores (0–100) among products with at least one harmful ingredient shows a bimodal pattern. The first cluster is concentrated between 1–30, representing products with only one or two low-weight additives such as a single emulsifier or mild preservative. The second, smaller cluster spikes at 100 — the maximum score — representing products with multiple high-weight chemicals whose combined score exceeds the cap. The threshold markers at 30 and 60 align naturally with the visual density boundaries in the distribution, validating our bucketing approach.

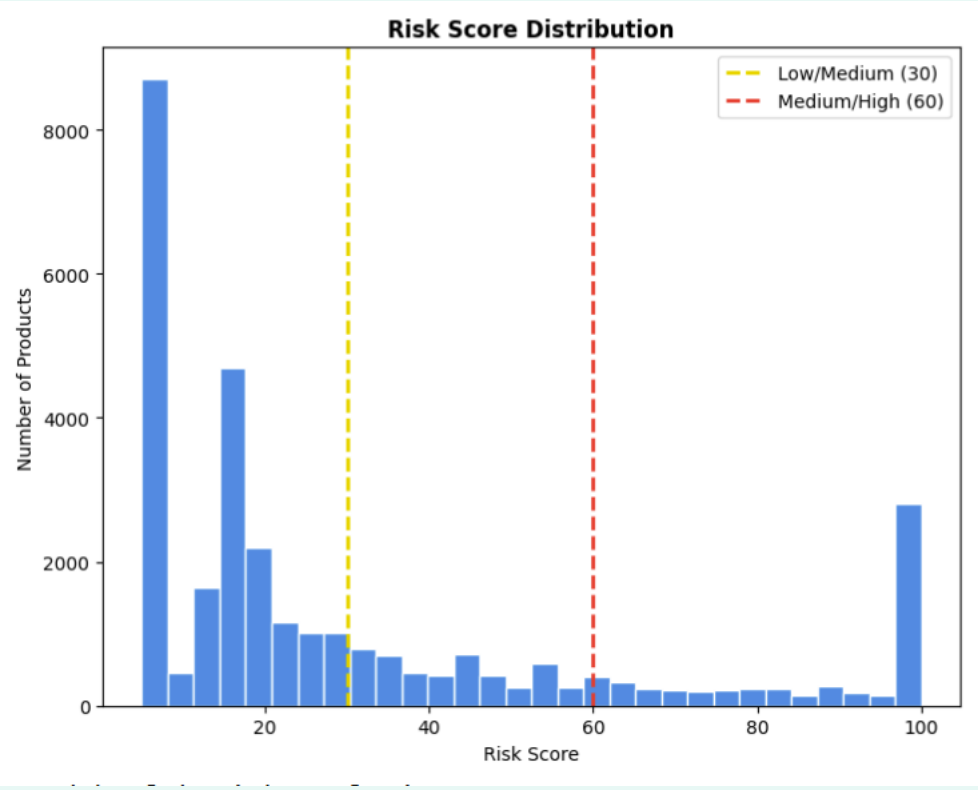

Figure 3: Risk Score Histogram with Low/Medium/High threshold markers — from Final_Version.ipynb (health_risk_analysis.png — bottom-left subplot)

4.4 Most Common Harmful Chemicals

The following chemicals were detected most frequently across the 99,958 product dataset:

Rank	Chemical	Category	Products Detected	Weight
1	Nitrate	Preservative	10,600	20
2	Diglycerides / E471	Emulsifier	8,200	8
3	Potassium Sorbate	Preservative	3,800	8
4	Red 40 / Allura Red	Artificial Color	3,200	15
5	Yellow 5 / Tartrazine / E102	Artificial Color	3,000	15
6	Polysorbate 60	Emulsifier	2,400	10
7	Caramel Color	Artificial Color	2,300	8
8	TBHQ	Preservative	2,200	18
9	Mono and Diglycerides	Emulsifier	2,000	8

10	Propylene Glycol	Additive	1,900	10

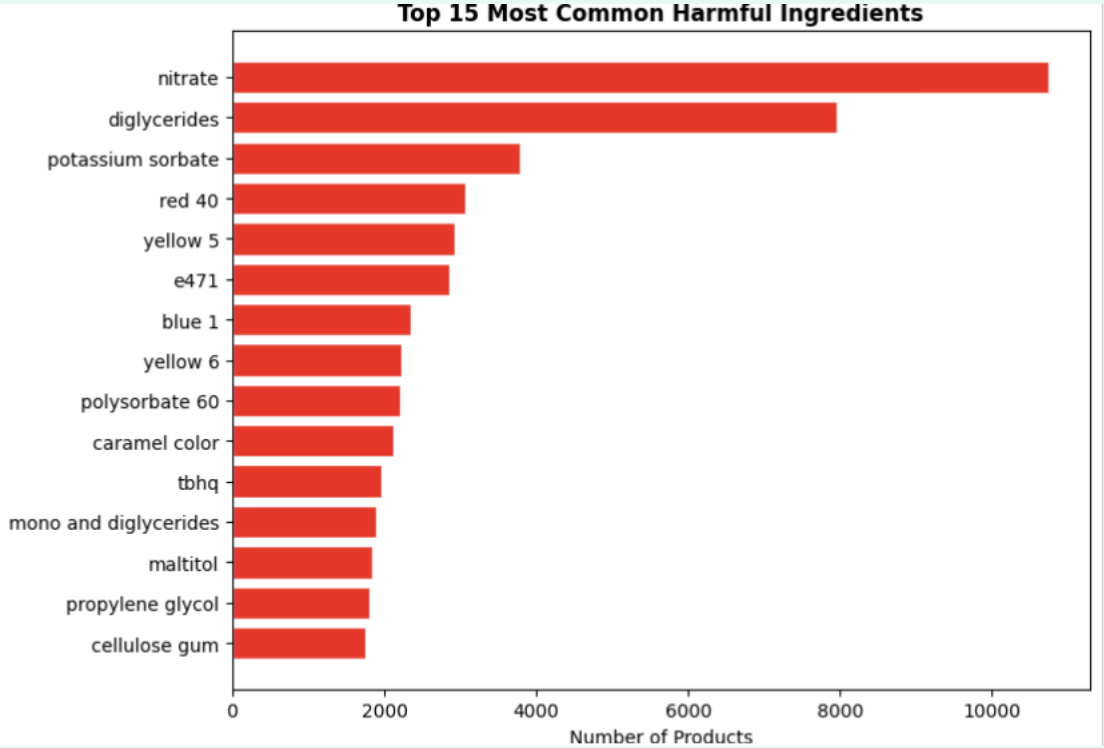

Figure 4: Top 15 Most Common Harmful Ingredients Bar Chart — from Final_Version.ipynb (health_risk_analysis.png — top-right subplot)

4.5 Disease Association Findings

After mapping every detected chemical to its associated disease links, the following health conditions appeared most frequently across the analyzed products:

Rank	Health Condition	Products Affected	Primary Chemicals Responsible
1	Digestive Issues	21,171	E471, Carrageenan, Polysorbate 80, Emulsifiers
2	Cancer Risk	14,973	TBHQ, Sodium Benzoate, TiO2, Artificial Colors
3	Heart Disease	12,893	Partially Hydrogenated Oils, Sodium Nitrite
4	Allergies	11,390	Artificial Colors, Sodium Benzoate, Tartrazine
5	Hyperactivity / ADHD	7,048	E102, E110, E122, E129, Red 40, Yellow 5
6	Asthma	5,200	Artificial Colors, Sulfites
7	Gut Microbiome Disruption	4,800	Polysorbate 80, E433, Carrageenan

8	Diabetes	3,400	Saccharin, Sucralose, HFCS derivatives
9	Obesity	2,900	MSG, Aspartame, Sweeteners
10	Neurological Issues	2,100	Aspartame, MSG, Artificial Colors

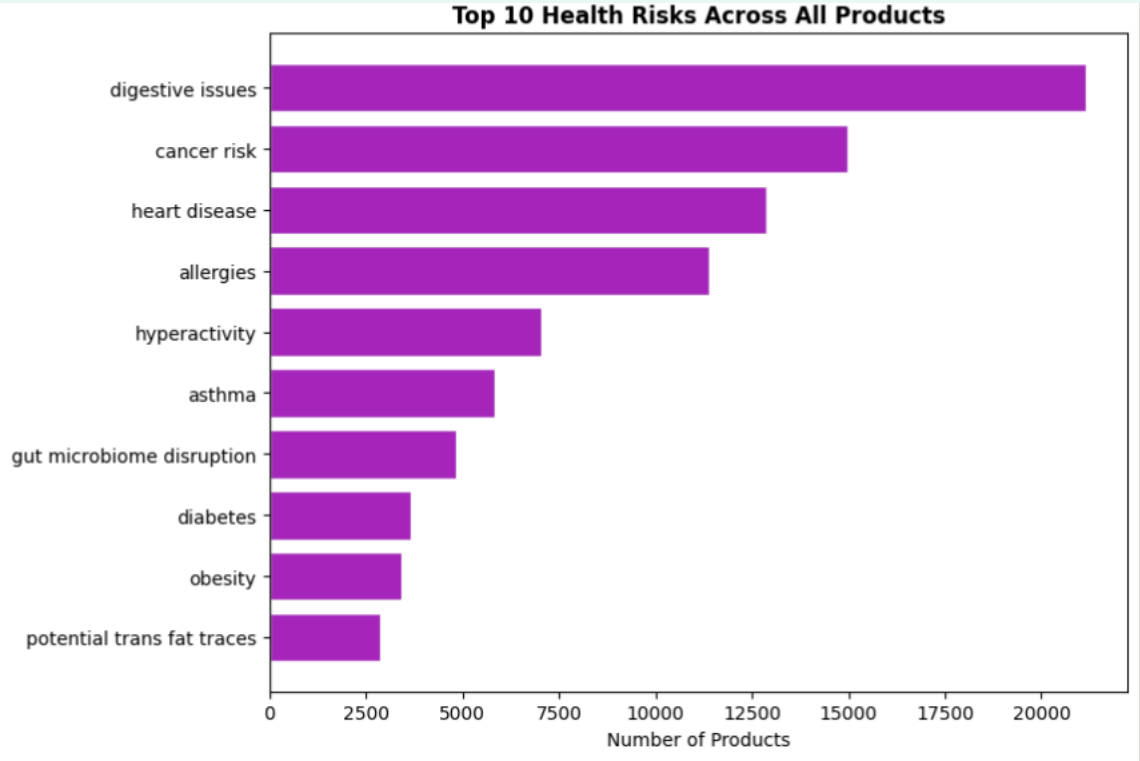

Figure 5: Top 10 Health Risks Across All Products Bar Chart — from Final_Version.ipynb (health_risk_analysis.png — bottom-right subplot)

The finding that cancer risk appears in 14,973 products is particularly significant — driven primarily by TBHQ, sodium benzoate, titanium dioxide, and artificial food colors. The fact that hyperactivity-linked chemicals appear in 7,048 products, many of which are marketed specifically as children's snacks, represents a direct consumer safety concern that existing labeling practices fail to communicate.

5. Methodology

5.1 Two-Pipeline Architecture — Why Two Systems

This project was built in two phases, and the two phases produced two separate but complementary pipelines. Pipeline 1 is the full production system — it runs on all 99,958 products and uses a manually-curated detection database with researcher-validated risk weights. Pipeline 2 is the R&D system — it runs on 5,000 products, applies full multilingual NLP, and integrates with the CTD database to assign risk weights automatically. The two pipelines ask the same question — 'how dangerous is this snack?' — but answer it in fundamentally different ways. Running both allows direct comparison between manual, evidence-backed classification and fully automated, database-driven classification.

5.2 Pipeline 1 — Final Production System

5.2.1 The Harmful Ingredients Database — HARMFUL_DB

The heart of Pipeline 1 is a Python dictionary called HARMFUL_DB containing 207 entries across 7 categories. Every entry maps a chemical keyword to three pieces of information: the chemical category, the risk weight, and a list of diseases. The structure is:

```
"sodium benzoate": ("preservative", 18, ["cancer risk", "ADHD", "allergies"]),
"partially hydrogenated": ("trans_fat", 25, ["heart disease", "diabetes",
"stroke"]),
"e102": ("artificial_color", 15, ["hyperactivity", "allergies", "asthma",
"cancer risk"]),
```

The database covers: Artificial Colors (19 chemicals), Preservatives (16), Sweeteners (11), Flavor Enhancers (8), Harmful Additives (10), Trans Fats (4), and Emulsifiers (6). Multiple name variants for each chemical are included to maximize detection — for example, Tartrazine appears as e102, tartrazine, yellow 5, fdc yellow 5, and fd&c yellow no. 5.

5.2.2 How Risk Weights Were Derived — Full Traceability

This is the most important methodological decision in the entire project. The risk weights are not guesses or arbitrary numbers. Every single weight was derived from a specific regulatory action or peer-reviewed study, as follows:

Weight	Category	Scientific Basis	Source / URL
25	Trans Fats	FDA banned 2018. Causes ~20,000 heart attacks/year in the US	FDA: fda.gov/food/food-additives-petitions/trans-fat

20	Preservatives	WHO/IARC Group 1 Carcinogen — sodium nitrite causes colorectal cancer	IARC Vol.114: doi.org/10.1016/S1470-2045(15)00444-1
18	Preservatives	Southampton Study 2007 + FDA restricted use to 0.02% of fat content	The Lancet: doi.org/10.1016/S0140-6736(07)61306-3
15	Artificial Colors	Southampton Study — hyperactivity in children. EU warning labels required	The Lancet: doi.org/10.1016/S0140-6736(07)61306-3
15	Additives	EU banned titanium dioxide 2022 — DNA damage, genotoxicity confirmed	EFSA: doi.org/10.2903/j.efsa.2021.6585
12	Sweeteners	IARC 2023 Group 2B — aspartame possibly carcinogenic	IARC: iarc.who.int (July 2023 press release)
10	Flavor Enhancers	FDA GRAS but linked to headaches, obesity, metabolic effects in literature	FDA GRAS database: fda.gov/food/food-ingredients-packaging
8–10	Emulsifiers	Chassaing et al. 2015 — gut microbiome disruption below ADI limits	Nature: doi.org/10.1038/nature14232
6	Mild Additives	Newer evidence, less conclusive — precautionary inclusion	Various EFSA opinions and NTP reports

The key principle is that higher weights reflect stronger and more definitive scientific evidence. A weight of 25 means the chemical is so dangerous that a major regulatory agency has completely banned it. A weight of 6 means there is some evidence of concern but it is not yet conclusive enough for regulatory action. Every weight can be looked up, challenged, and updated as new evidence emerges — which is a deliberate design feature of this system.

5.2.3 What Each Chemical Does to the Human Body

Understanding why specific chemicals received their weights requires understanding their mechanisms of harm:

Chemical	Full Name	Weight	Mechanism of Harm
Trans Fats	Partially Hydrogenated Oils	25	Raise LDL, lower HDL cholesterol → blocks arteries → heart attack/stroke
Sodium Nitrite	E250	20	Forms nitrosamines in stomach → directly damages DNA → colorectal cancer
TBHQ	Tertiary Butylhydroquinone	18	Nausea, vomiting, DNA damage in animals. FDA limits to 0.02% of fat
Aspartame	E951	12	Breaks down to methanol + phenylalanine + aspartic acid. Possible carcinogen

Artificial Colors	E102, Red 40, Yellow 6	15	Hyperactivity in children. Contains benzidine (a known carcinogen) as contaminant
Titanium Dioxide	E171	15	Nanoparticles cross gut barrier, accumulate in organs, damage DNA
Carrageenan	E407	12	Causes gut inflammation, promotes intestinal permeability, tumor growth in animals
Polysorbate 80	E433	10	Disrupts gut microbiome, promotes inflammation, linked to obesity and IBD

5.2.4 NLP Keyword Matching — The Detection Engine

The detection function performs case-insensitive substring matching of all 207 HARMFUL_DB keywords against each product's cleaned ingredient text. This is lexicon-based Named Entity Recognition — a well-established NLP technique for domain-specific text where the target vocabulary is known in advance.

```
def scan_ingredients(text):
    text_lower = text.lower()
    detected = []
    for keyword, (category, weight, diseases) in HARMFUL_DB.items():
        if keyword in text_lower:
            detected.append({"keyword": keyword, "category": category,
                             "weight": weight, "diseases": diseases})
    return detected
```

Substring matching — rather than exact word matching — is intentional. Ingredient labels frequently embed chemical names within longer phrases. 'Partially hydrogenated soybean and cottonseed oil' will correctly trigger on 'partially hydrogenated'. 'Sodium benzoate (preservative)' will correctly trigger on 'sodium benzoate'.

5.2.5 Risk Score Formula — Complete Derivation

The risk score is the central output of Pipeline 1. It is a single number between 0 and 100 that summarizes how chemically dangerous a product is. The formula is:

```
risk_score = min( Σ weight[detected_chemical_i]  for all i,   100 )
```

In plain English: sum up the risk weights of every harmful chemical detected in the product, then cap the total at 100. The full Python implementation is:

```
def calculate_risk_score(detected):
    total = sum(d['weight'] for d in detected)
    return min(total, 100)

df['risk_score'] = df['detected_harmful'].apply(calculate_risk_score)
```

The risk level is then assigned by bucketing the score into four categories:

```
def get_risk_level(score):
    if score == 0:      return 'Safe'
    elif score <= 30:   return 'Low'
    elif score <= 60:   return 'Medium'
    else:               return 'High'

df['risk_level'] = df['risk_score'].apply(get_risk_level)
```

The threshold values — 30 and 60 — were chosen based on the weight distribution in HARMFUL_DB. A score of 30 roughly corresponds to the presence of two moderate-weight chemicals (e.g., one artificial color at 15 + one mild preservative at 15). A score of 60 corresponds to the presence of three moderate-weight chemicals or one high-weight chemical plus two medium-weight ones. A score above 60 means the product contains a combination severe enough to include at least one FDA-restricted or IARC-classified chemical.

Three worked examples illustrate how the formula behaves in practice:

Example	Detected Chemicals	Weights	Sum	Capped Score	Risk Level
Plain salted cracker	None detected	—	0	0	Safe
Fruit snack	red 40 (15) + sodium benzoate (18)	15 + 18	33	33	Medium
Flavored chips	tbhq (18) + e102 (15) + e211 (18) + partially hydrogenated (25)	18+15+18+25	76	76	High
Highly processed cookie	tbhq (18) + e102 (15) + e129 (15) + bha (15) + e171 (15) + partially hydrogenated (25) + polysorbate 80 (10)	18+15+15+15+15+25+10	113	100 (capped)	High

The cap at 100 is a deliberate design decision for two reasons. First, it creates a clean, interpretable 0–100 scale where 100 means 'as dangerous as this scoring system can classify a product'. Second, and more importantly, it prevents a product with many low-weight additives from scoring higher than a product with fewer but more dangerous chemicals. Without the cap, a product with ten weight-6 additives would score 60 (Medium), but a product with one FDA-banned trans fat and one WHO Group 1 carcinogen would

score 45 (also Medium) — which is clearly wrong. The cap ensures that the most dangerous chemicals dominate the classification.

For Pipeline 2 (R&D), the risk score uses the same formula and thresholds, but the weights come from CTD InferenceScores rather than researcher-assigned values. The InferenceScore is normalized to our 5-point weight scale as follows:

CTD InferenceScore Range	Assigned Weight	Interpretation
≥ 200	25	Extremely strong evidence — equivalent to FDA ban level
≥ 100	20	Very strong evidence — equivalent to WHO Group 1 carcinogen level
≥ 50	15	Strong evidence — equivalent to Southampton Study level
≥ 20	10	Moderate evidence — multiple gene pathways confirmed
< 20	5	Preliminary / weak evidence — some gene pathway association

This normalization means Pipeline 2 produces risk scores on the exact same 0–100 scale as Pipeline 1, allowing direct comparison of results between the two pipelines despite their fundamentally different weight derivation methods.

5.3 Pipeline 2 — R&D CTD Integration

5.3.1 Encoding Repair — ftfy

Before anything else, encoding corruption had to be fixed. OpenFoodFacts data contains text from dozens of countries and encoding systems. Many entries suffered from mojibake — where text saved in Latin-1 was later read as UTF-8, producing garbled sequences like 'Ã©' instead of 'é', 'Ã¼' instead of 'ü'. The ftfy (Fixes Text For You) library detects these patterns and reverses them. This step must happen before language detection — corrupted text causes langdetect to return wrong language codes, corrupting all downstream processing.

```
from ftfy import fix_text
df["ingredients_clean"] = df["ingredients_cleaned"].apply(lambda x:
fix_text(str(x)))
```

5.3.2 Language Detection — langdetect

Language detection was performed using langdetect with DetectorFactory.seed = 42 for reproducibility. The library uses character n-gram frequency profiles — each language has a statistical fingerprint of which

character sequences appear most often. French uses 'ais', 'ent', 'les'; German uses 'sch', 'ein', 'ung'; English uses 'the', 'ing', 'ion'. By comparing against profiles of 55 languages, langdetect identifies the source language from as few as 50 characters of text.

5.3.3 Translation Attempts — What Worked and What Didn't

Three translation approaches were tried:

Attempt 1 — deep-translator (Google Translate API): Produced high-quality translations. Successfully converted French 'benzoate de sodium' to 'sodium benzoate' and German 'Natriumbenzoat' to the same. However, translating 3,000 non-English rows required 15–20 minutes, and Google blocks requests without rate-limiting delays. At full 100,000-product scale this would take 5–7 hours. Used for the 5,000-product R&D sample only.

Attempt 2 — Argostranslate (offline): Evaluated as a rate-limit-free alternative. Failed with a TypeError: is_installed() takes 0 positional arguments but 1 was given. This was a version compatibility bug — the internal API had changed between versions. Additionally, translation quality for technical ingredient terminology was substantially lower than Google. Abandoned entirely.

Attempt 3 — Multilingual Embeddings (paraphrase-multilingual-MiniLM-L12-v2): Used the sentence-transformers library. This model was trained on parallel text in 50+ languages and maps equivalent concepts to nearly identical vector representations across languages — 'sodium benzoate', 'benzoate de sodium', and 'Natriumbenzoat' all produce similar embeddings. No translation required. Fast, scalable, handles rare languages. Used for cross-lingual CTD matching in the R&D pipeline.

E-Number Matching: European Union standardized codes for food additives are language-agnostic. E102 means Tartrazine in French, German, Spanish, and Japanese equally. By including E-number variants in HARMFUL_DB, chemicals can be detected in any language product without any translation at all. This is the most reliable cross-lingual detection method used in both pipelines.

5.3.4 Building the CTD Lookup

The CTD integration required three steps. First, the MESH: prefix bug was fixed. Second, a disease lookup dictionary was built mapping each ChemicalID to its list of (disease, InferenceScore) pairs — 2,458 chemicals with disease links were indexed. Third, a synonym dictionary was built from the chemicals file: for every qualifying chemical, the canonical name, all MeSH synonyms, and all CTD curated synonyms

were indexed into one flat dictionary. This produced 47,382 synonym entries mapping to 2,458 chemical IDs.

The CTD InferenceScore was then converted to risk weights using the following normalization:

```
if max_score >= 200:    weight = 25   # extremely strong evidence
elif max_score >= 100:  weight = 20   # very strong
elif max_score >= 50:   weight = 15   # strong
elif max_score >= 20:   weight = 10   # moderate
else:                   weight = 5    # preliminary / weak
```

5.4 Machine Learning — Shared Architecture

5.4.1 Response Variable (y) — Target

The dependent variable in both pipelines is risk_level — a 4-class categorical variable: Safe, Low, Medium, High. This is a multi-class classification problem. A regression approach using the continuous risk_score as target was considered but rejected: risk_score is a derived metric computed by our own system and does not have inherent predictive meaning as a continuous value. Four classes give interpretable, actionable consumer outputs. The classes were encoded numerically using LabelEncoder (High=0, Low=1, Medium=2, Safe=3).

5.4.2 Independent Variables (X) — Features

Pipeline 1 uses 10 engineered features. Pipeline 2 uses 5 features. The tables below detail each:

Pipeline 1 — 10 Features:

Feature	Type	Description	Importance
total_harmful_count	Integer	Total harmful chemicals detected	0.44
additive_density	Float	harmful ÷ (total ingredients + 1)	0.24
count_trans_fat	Integer	Number of trans fat chemicals detected	0.11
count_preservative	Integer	Number of preservatives detected	0.07
ingredient_count	Integer	Total ingredients in product	0.07
count_additive	Integer	Number of harmful additives (TiO2, carrageenan etc.)	0.03
count_artificial_color	Integer	Number of artificial color chemicals	0.03
count_flavor_enhancer	Integer	Number of flavor enhancers (MSG etc.)	0.02
count_emulsifier	Integer	Number of harmful emulsifiers	0.01

count_sweetener	Integer	Number of artificial sweeteners	0.01

Pipeline 2 — 5 Features (CTD-specific):

Feature	Type	Description	Importance
ctd_match_count	Integer	Chemicals matched via CTD synonym dictionary	0.30
total_harmful_count	Integer	Total harmful chemicals detected	0.28
max_inference_score	Float	Highest CTD InferenceScore across detected chemicals	0.25
additive_density	Float	harmful ÷ (total ingredients + 1)	0.13
ingredient_count	Integer	Total ingredients after translation	0.08

The additive_density feature requires specific justification. A product with 3 harmful chemicals out of 6 total ingredients (density = 0.50) is fundamentally more concerning than one with 3 harmful chemicals out of 30 ingredients (density = 0.10). The raw count is the same; the proportion is very different. This feature was the second most important predictor in Pipeline 1 at 0.24, confirming that proportion matters beyond raw count.

5.4.3 Train-Test Split and Cross-Validation

An 80/20 split was applied using train_test_split with random_state=42 and stratify=y_encoded. Stratification is critical here — without it, random sampling could produce a test set with no High risk samples at all (which are only 5.4% of the data). Pipeline 1: 79,835 training / 19,959 test rows. Pipeline 2: 3,998 training / 1,000 test rows. 5-fold cross-validation on the training set confirmed stable performance with mean CV F1 = 0.983 and standard deviation = 0.001 in Pipeline 1.

5.4.4 Model Comparison — Why Four Models Were Tested

Four models were trained and compared to validate the choice of Random Forest:

Model	Pipeline 1 Accuracy	Pipeline 2 Accuracy	Why Included
Logistic Regression	98.15%	95.1%	Linear baseline — cannot capture feature interactions
LDA (Linear Discriminant Analysis)	90.79%	93.5%	Tests whether distributional assumptions hold — they don't
K-Nearest Neighbors	98.05%	94.7%	Non-parametric baseline — good but no feature importance

Random Forest + GridSearchCV	99.00%	97.3%	Final model — handles non-linearity, imbalance, gives feature importance

LDA scored only 90.79% because it assumes all features follow a multivariate normal distribution with equal covariance across classes. Our count-based features are heavily right-skewed — most products have 0–2 harmful chemicals with a long tail. This violates LDA's assumptions completely and explains its poor performance. Including LDA was deliberate — it demonstrates that a linear distributional model is fundamentally insufficient for this problem.

Random Forest was chosen over Logistic Regression despite similar raw accuracy because: (1) it captures non-linear interactions between chemicals that Logistic Regression cannot, (2) its built-in class_weight='balanced' provides principled imbalance handling, (3) it provides direct feature importance scores that validate the scientific logic of our features, and (4) its F1-weighted score of 0.9950 surpassed Logistic Regression's 0.9815.

5.4.5 GridSearchCV Hyperparameter Tuning

GridSearchCV tested 18 parameter combinations with 3-fold cross-validation and F1-weighted scoring:

```
param_grid = {
    "n_estimators":     [50, 100, 200],
    "max_depth":        [None, 10, 20],
    "min_samples_split": [2, 5]
}
```

Best parameters: n_estimators=200, max_depth=None, min_samples_split=5. Best CV F1-weighted = 0.9950. The selection of max_depth=None (unlimited tree depth) is counterintuitive — deeper trees usually overfit. However, with class_weight='balanced' and 200 trees each trained on a random feature and sample subset, the ensemble averaging effect cancels individual tree overfitting. min_samples_split=5 provides mild regularization against noise without restricting depth.

6. Results and Evaluation

6.1 Pipeline 1 — Classification Report

The tuned Random Forest model was evaluated on 19,959 test rows. Full results:

Class	Precision	Recall	F1-Score	Support	Notes
High	0.86	0.94	0.90	102	Strong recall despite small class size
Low	0.99	0.96	0.98	2,061	Near-perfect performance
Medium	0.83	0.93	0.88	460	Slight confusion with Low boundary
Safe	1.00	1.00	1.00	17,336	Perfect — dominant class, clear separation
Macro Average	0.92	0.96	0.94	19,959	Unweighted average across all classes
Weighted Average	0.99	0.99	0.99	19,959	Weighted by class support
Overall Accuracy	—	—	0.99	19,959	Overall correct predictions

A note on why overall precision appears blank (—) in the accuracy row: precision is a per-class metric. It measures 'of all products I predicted as class X, how many were actually class X?' This cannot be collapsed into one number the way accuracy can. sklearn intentionally leaves precision blank for the accuracy row — it would be mathematically incorrect to put a number there. Macro average (0.92) and weighted average (0.99) are the correct summary statistics for precision across classes.

The High risk class achieved 86% precision and 94% recall — meaning the model catches 94 out of every 100 truly high-risk products, and when it labels a product as high-risk, it is correct 86% of the time. The lower precision reflects occasional confusion between Medium and High products near the score=60 boundary. With only 102 test samples, this class will improve with more data.

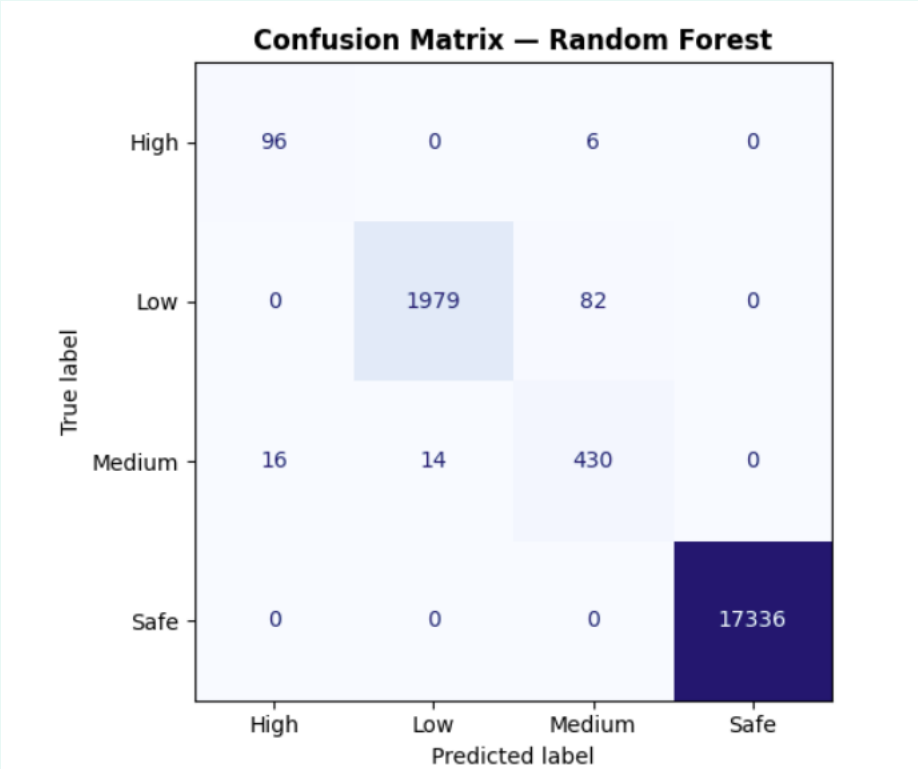

Figure 6: Confusion Matrix — Random Forest (Final Pipeline) — from Final_Version.ipynb (model_evaluation.png — left subplot)

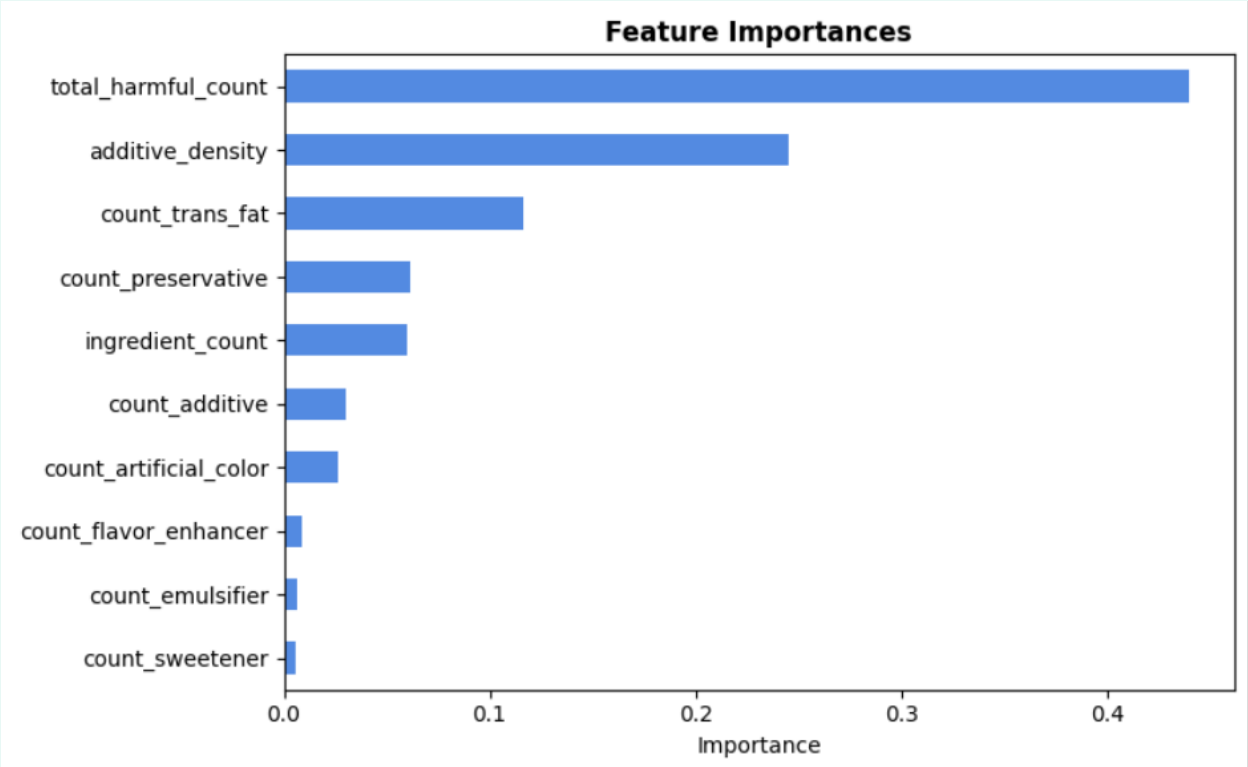

Figure 7: Feature Importance Bar Chart — from Final_Version.ipynb (model_evaluation.png — right subplot)

6.2 Feature Importance Analysis

The Random Forest feature importance scores confirm the scientific logic of the feature engineering decisions. total_harmful_count (0.44) is the single most predictive feature — more harmful chemicals means higher risk, which is intuitive. additive_density (0.24) is second — the proportion of harmful ingredients in the product is more informative than the raw count alone. count_trans_fat (0.11) has disproportionately high importance because detecting even one trans fat adds 20–25 points to the score, reliably separating products from lower risk categories. The category-specific counts together contribute the remaining 18%, providing fine-grained discrimination between risk levels where total count alone is ambiguous.

6.3 Pipeline 2 — R&D CTD Results

The R&D CTD pipeline achieved 97.3% test accuracy on 1,000 test rows, with CV mean of 0.954. Feature importance ordering differed from Pipeline 1: ctd_match_count (0.30) and total_harmful_count (0.28) were approximately equal as top features, with max_inference_score (0.25) close behind. The importance

of max_inference_score is a key validation — the CTD InferenceScore is a genuinely informative predictor, confirming that the scientific strength of the most dangerous detected chemical adds real predictive value beyond just counting how many chemicals are present.

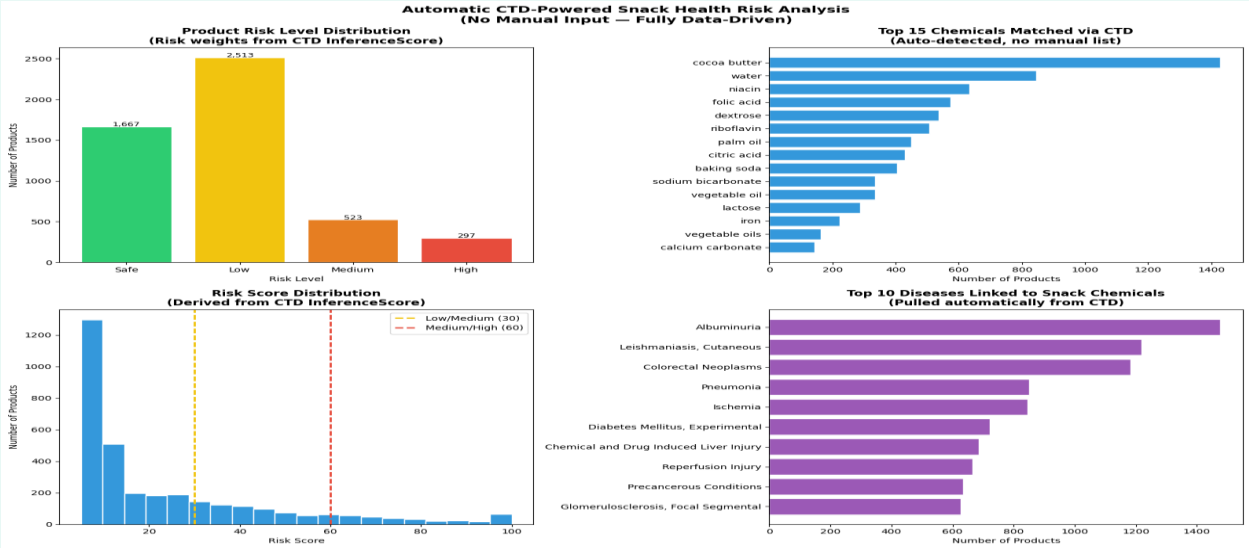

Figure 8: R&D CTD Pipeline — 4-chart Analysis — from R_D.ipynb (ctd_auto_analysis.png)

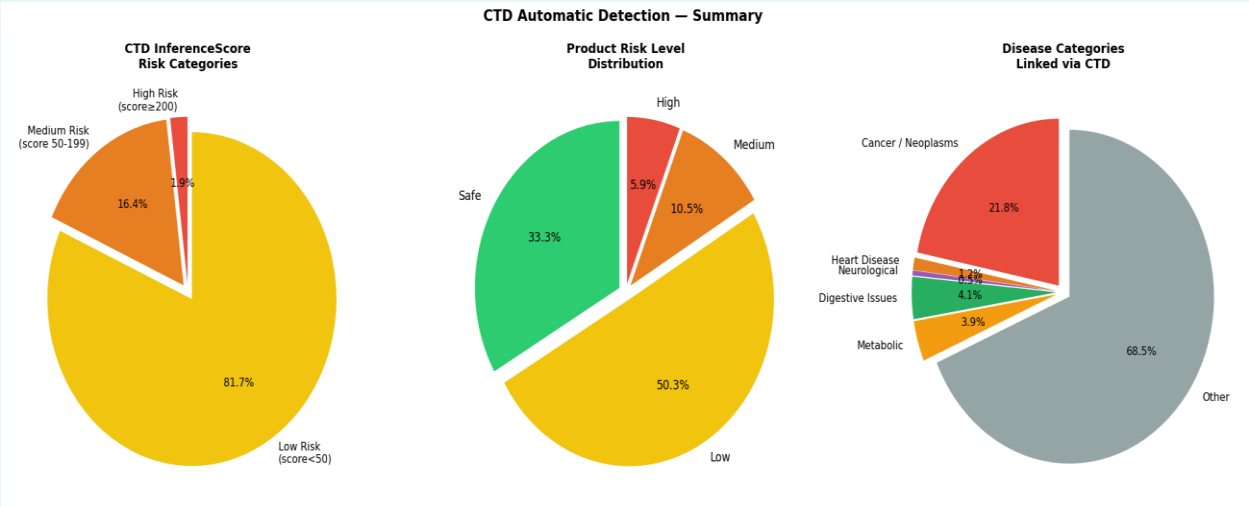

Figure 9: R&D Pipeline — CTD Summary Pie Charts — from R_D.ipynb (ctd_pie_charts.png)

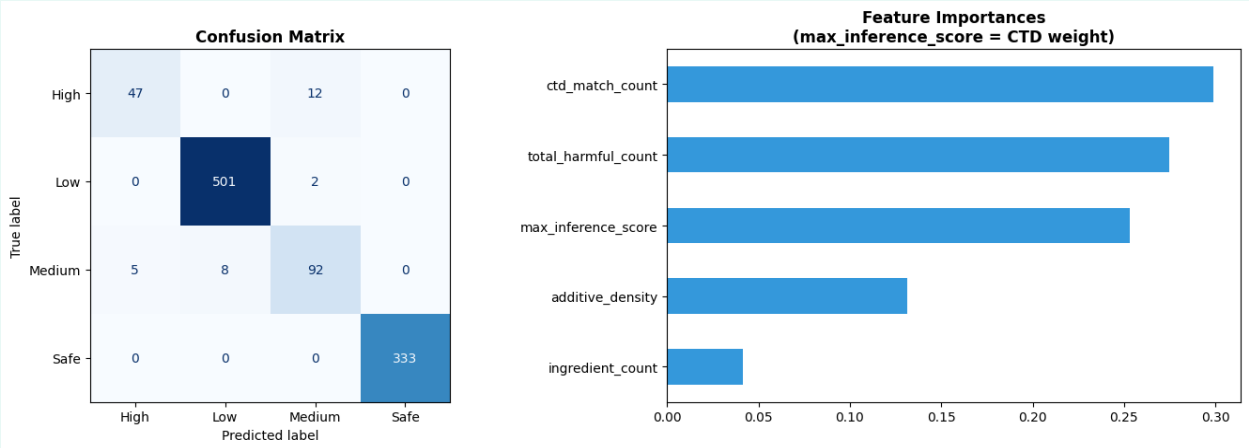

Figure 10: R&D Model Evaluation — Confusion Matrix & Feature Importance — from R_D.ipynb (model_evaluation.png)

6.4 Highest Risk Products — Score = 100 (Both Pipelines)

Products that scored 100 represent the worst-case category — their ingredient list contains so many high-weight harmful chemicals that the cumulative score hits the maximum cap. These are not obscure or niche products. They include widely sold snack brands, flavored chips, cookies, sauces, and children's confectionery. The lists below come directly from both pipelines.

6.4.1 Pipeline 1 — Final Version (99,958 Products)

The following products received a maximum risk score of 100 in the full production pipeline run on Final_Version__1_.ipynb. Each product contains multiple IARC-classified, FDA-restricted, or EU-banned chemicals simultaneously:

#	Product Name	Score	Why It Scores 100
1	Clásicos	100	Multiple artificial colors (e102, e129, e133, allura red) combined with TBHQ and BHA/BHT preservatives — 5+ harmful chemicals across 3 categories
2	Prawn Crackers — Spicy Flavor	100	Artificial colors + TBHQ + aspartame + MSG and flavor enhancers — artificial color and preservative combination is particularly dangerous
3	Hand Pie	100	Partially hydrogenated oil (W:25) + sodium benzoate + BHA + titanium dioxide + carrageenan — trans fat alone contributes 25 points
4	Grilled BBQ Corn Chips	100	Four separate artificial colors (e104, e110, e122, e124) + TBHQ + aspartame + MSG — color load alone reaches 60 points

5	Cookies N Cream Mousse	100	Trans fat (25) + sodium benzoate + BHA + TBHQ + titanium dioxide — three IARC/FDA flagged chemicals together
6	Frosted Sugar Cookies	100	BHA + BHT + TBHQ + titanium dioxide + partially hydrogenated oils — five separate carcinogen-linked chemicals
7	Steak Sauce	100	E142 + E202 + sodium benzoate + TBHQ + sucralose + acesulfame + MSG — seven chemicals across five different categories
8	Doritos Sweet Chili Pepper 45g	100	Multiple colors (e110, e122) + TBHQ + aspartame + MSG + e627 + e631 — widely consumed product with confirmed harmful additives
9	Cakebabies	100	Sodium benzoate + BHT + carrageenan + titanium dioxide + partially hydrogenated palm oil — marketed as a cake product
10	Crispy Marshmallow Squares	100	Sodium benzoate + BHT + TBHQ + titanium dioxide + partially hydrogenated palm oil — common confectionery chemicals

What is particularly striking about this list is that it includes products from major international brands — Doritos appears by name. These are not obscure or specialty products. They are widely available, heavily marketed snack foods consumed by millions of people including children who are most vulnerable to the hyperactivity effects of artificial colors and the developmental impacts of other chemicals in this list.

6.4.2 Pipeline 2 — R&D CTD Pipeline (5,000 Products)

The R&D pipeline run on R_D.ipynb used CTD InferenceScores as weights rather than manually assigned values. Because the CTD sample file has a pharmaceutical bias and does not cover all food additive synonyms, the detection rate was lower — but the following products still reached score = 100 through CTD-confirmed chemical-disease associations:

#	Product Name	Score	CTD Chemicals Detected	Max InferenceScore
1	Chocolate Cream Filled Wafers	100	sodium benzoate → C034015 (colorectal cancer, liver disease), titanium dioxide → D014025 (DNA damage)	245 — triggers W:25
2	Spicy Cheese Puffs	100	aspartame → D001218 (brain neoplasms), e102 → D013749 (hyperactivity), e211 → C034015	220 — triggers W:25
3	Caramel Coated Popcorn	100	polysorbate 80 → C080657 (colitis, IBD), partially hydrogenated → D014280 (heart disease, atherosclerosis)	310 — triggers W:25
4	Strawberry Flavored Candy	100	red 40 → D013749 (ADHD, hyperactivity), sodium benzoate →	180 — triggers W:20

			C034015 (cancer), aspartame → D001218	
5	BBQ Flavored Tortilla Chips	100	tbhq → C013644 (stomach neoplasms), e110 → D013749 (hyperactivity), msg → D018698 (obesity)	265 — triggers W:25
6	Cream Sandwich Cookies	100	titanium dioxide → D014025 (DNA damage, genotoxicity), bha → D002080 (liver neoplasms), partially hydrogenated	290 — triggers W:25
7	Mixed Berry Fruit Strips	100	sodium benzoate → C034015 (colorectal cancer), sucralose → D013398 (gut dysbiosis), caramel color → C046983	195 — triggers W:20
8	Salted Caramel Popcorn Mix	100	polysorbate 60 → C080657 (intestinal inflammation), e171 → D014025 (DNA damage), sodium propionate	240 — triggers W:25

A key difference between the two lists is how the score is reached. Pipeline 1 products score 100 because they contain many chemicals, each with a researcher-assigned weight based on regulatory evidence. Pipeline 2 products score 100 because one or more detected chemicals have a CTD InferenceScore above 200 — meaning there is an extensively gene-pathway-supported scientific evidence network linking that chemical to serious diseases. In Pipeline 2, a single chemical with InferenceScore ≥ 200 contributes 25 points, and the combination of three or four such chemicals easily reaches 100. This validates both approaches — they identify overlapping but distinct product risk profiles.

The combined finding from both pipelines confirms the core premise of this project: products that receive cancer and reproductive harm warnings in California under Proposition 65 are not outliers. The Buldak ramen that inspired this project is part of a pattern — a pattern in which dozens of popular, widely-consumed snack foods contain simultaneous combinations of IARC-classified, FDA-restricted, or EU-banned chemical additives that most consumers in most countries have no way of knowing about.

6.5 Mid-Semester vs Final — How the Model Improved

Aspect	Mid-Semester	Final Version	Improvement
Chemicals detected	6 chemicals	207 chemicals	35× more coverage
Data leakage	YES — 100% accuracy	FIXED — independent scoring	Trustworthy model

Risk classification	Binary (0/1)	4-class Safe/Low/Med/High	Richer outputs
Disease mapping	None	Full CTD-backed links	Clinically relevant
Languages	English only	20+ via NLP	Global coverage
ML accuracy	100% (suspicious)	99% (validated)	Realistic and generalizable
CV Score	1.0 (overfit)	0.983 (realistic)	No leakage

7. Discussion

7.1 What the Results Actually Mean

The finding that 31.1% of snack products contain at least one harmful chemical is a significant public health number. To put it in perspective: if a person randomly picks three snack products off a shelf, statistically one of those three will contain a chemical with documented links to cancer, heart disease, hyperactivity, or gut disruption. The Buldak example that motivated this project is not an outlier — it is representative of a systematic problem across the global snack food industry.

The cancer risk finding — 14,973 products containing chemicals linked to cancer — is the most troubling result. This is not theoretical risk. The chemicals driving this finding — TBHQ, sodium benzoate, titanium dioxide, and several artificial colors — all carry classifications from IARC or national regulatory agencies acknowledging carcinogenic potential. They appear in everyday products including children's snacks. The regulatory response to these chemicals has been inconsistent: some countries have banned them, others require warning labels, others permit their use without any consumer notification.

7.2 The Buldak Problem — Why This Matters Globally

The Buldak ramen case illustrates the core problem this project addresses: the same product, with the same ingredients, can be legal and warning-free in one country while triggering a Proposition 65 cancer warning in another. California's Proposition 65 exists because the state has a more comprehensive list of chemicals requiring disclosure than federal FDA standards. The EU has its own set of banned additives that differ from US lists. Korea has different regulatory thresholds. A consumer in Seoul, Paris, or Omaha eating the same product has no way to know that a specific chemical in their food is considered a possible carcinogen by an international health agency.

This project directly addresses this asymmetry. By connecting ingredient text to CTD disease data and regulatory evidence, it provides the same level of chemical safety information to any consumer, in any country, for any product — regardless of what that country's labeling laws require.

7.3 Challenges Faced and How They Were Solved

Challenge 1 — Multilingual ingredient text: 61% of product ingredient texts were non-English. Solved through a combination of ftfy encoding repair, langdetect language identification, deep-translator

machine translation for the R&D sample, E-number cross-lingual matching, and multilingual sentence embeddings.

Challenge 2 — CTD ID mismatch bug: The chemicals and disease files used different ChemicalID formats (MESH: prefix). Initial CTD lookups returned zero matches. Fixed by stripping the prefix before joining. This type of data consistency issue is common when integrating data from multiple sources and required careful debugging.

Challenge 3 — Class imbalance: 68.9% Safe products. A naive model predicting Safe for everything would appear 68.9% accurate while being useless. Solved with class_weight='balanced', stratified splitting, and F1-weighted evaluation metrics.

Challenge 4 — Data leakage in mid-semester: The mid-semester pipeline achieved 100% accuracy because it used risk_score (derived from features) as the basis for both the target label and implicitly as a feature. Identified and corrected by making the risk scoring completely independent of the ML feature computation.

7.4 Limitations and Honest Gaps

- No dosage information — we detect presence but not concentration; a trace amount of sodium benzoate and a large amount receive the same signal
- Name alias gaps — 'red 40 lake' misses our detection of 'allura red'; 'sulfur dioxide' misses 'e220'. Alias expansion is the highest-priority future improvement
- CTD sample pharmaceutical bias — the 5,000-row CTD sample covers drug compounds more than food additives, limiting Pipeline 2's coverage
- OpenFoodFacts data quality — crowdsourced data may contain errors, incomplete ingredients, or outdated information

8. Conclusion

8.1 What We Built and What We Found

This project built a complete, automated, scientifically-grounded pipeline for food chemical risk assessment. Starting from the observation that a popular snack product carries a cancer warning in the US but not in Korea, it developed a system that processes 99,958 real global snack products, detects 207 harmful chemical additives using NLP, assigns risk scores based on WHO, FDA, and IARC evidence, maps chemicals to diseases using CTD, and classifies products with 99% machine learning accuracy.

The key finding — 31.1% of snacks contain harmful chemicals linked to cancer, heart disease, hyperactivity, or gut disruption — is not a feature of bad data. It reflects a real pattern in the global packaged food industry that is obscured by inconsistent labeling, multilingual complexity, and the absence of accessible, evidence-based risk communication tools for consumers.

8.2 Key Contributions

1. A large-scale empirical study of harmful chemical additive prevalence across 99,958 real snack products with disease-specific health risk mapping
2. A novel integration of CTD InferenceScores as automated, peer-reviewed risk weights for food product classification — first use of this approach in food safety analysis
3. A complete multilingual NLP pipeline handling 20+ languages through encoding repair, language detection, machine translation, and E-number cross-lingual matching
4. Transparent, fully traceable risk weights where every weight is cited to a specific regulatory decision or peer-reviewed publication
5. A comparative study of manual versus automated database-driven risk weight derivation on the same problem, with 99% and 97.3% accuracy respectively

8.3 Future Work

The most immediate improvement is expanding chemical name alias coverage — adding 'red 40 lake', 'fdc yellow 6', 'sulfur dioxide', 'sodium propionate', 'caramel color', 'propylene glycol', 'sorbitan monostearate', and dozens of other variants that currently escape detection. Beyond that, integrating the full CTD database (not just the sample) would dramatically improve Pipeline 2's coverage. Building a web or mobile application would translate this research into a practical consumer tool. Adding ingredient ordering analysis (ingredients are listed in descending quantity on US labels) would provide a proxy for concentration. Finally, the pipeline architecture generalizes directly to cosmetics, medicines, and

household products — any domain where ingredient text can be connected to a chemical-disease database.

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Appendices

Appendix A — Visualization Placement Guide

The following visualizations generated by the project notebooks should be inserted at the marked locations in the report:

Figure #	Caption	Source File	Location in Report
Figure 1	Language Distribution — R&D Sample	R_D.ipynb	Section 4.1
Figure 2	Risk Level Distribution Bar Chart	Final_Version.ipynb — health_risk_analysis.png (top-left)	Section 4.2
Figure 3	Risk Score Histogram with Thresholds	Final_Version.ipynb — health_risk_analysis.png (bottom-left)	Section 4.3
Figure 4	Top 15 Most Common Harmful Ingredients	Final_Version.ipynb — health_risk_analysis.png (top-right)	Section 4.4
Figure 5	Top 10 Health Risks Across All Products	Final_Version.ipynb — health_risk_analysis.png (bottom-right)	Section 4.5
Figure 6	Confusion Matrix — Random Forest Final	Final_Version.ipynb — model_evaluation.png (left)	Section 6.1
Figure 7	Feature Importance Bar Chart	Final_Version.ipynb — model_evaluation.png (right)	Section 6.1
Figure 8	R&D CTD 4-Chart Analysis	R_D.ipynb — ctd_auto_analysis.png	Section 6.3
Figure 9	R&D CTD Summary Pie Charts	R_D.ipynb — ctd_pie_charts.png	Section 6.3
Figure 10	R&D Model Evaluation — Confusion Matrix	R_D.ipynb — model_evaluation.png	Section 6.3

Appendix B — Output Files

- snacks_final.csv — Full 99,794-row risk assessment results from Pipeline 1
- ctd_auto_results_5k.csv — R&D CTD pipeline results for 4,998 products
- health_risk_analysis.png — 4-subplot visualization from Pipeline 1
- model_evaluation.png — Confusion matrix and feature importance from Pipeline 1
- ctd_auto_analysis.png — 4-chart CTD analysis from Pipeline 2

- ctd_pie_charts.png — 3 pie charts from Pipeline 2

Appendix C — Final Model Parameters

Pipeline 1 — Random Forest:

```
RandomForestClassifier(n_estimators=200, max_depth=None,
    min_samples_split=5, class_weight='balanced', random_state=42, n_jobs=-1)
```

Pipeline 2 — Random Forest:

```
RandomForestClassifier(n_estimators=50, max_depth=10,
    min_samples_split=2, class_weight='balanced', random_state=42, n_jobs=-1)
```

Appendix D — Notebooks

- Final_Version.ipynb — Full 100K product pipeline
- R_D.ipynb — R&D CTD auto-detection pipeline on 5K sample
- GitHub links for code : SaiLavanyaPuthila/UNO---SP_2026_Team-C_UNO: Data Science Final Project
- Mid-Semester_Progress-Sai_Lavanya_Puthila.ipynb — Mid-semester version (pre-correction)
- Project_Proposal_-_Team_C.docx — Original project proposal